

Business Analytics Terminology

Foundational Concepts of Analytics

① Business Analytics are statistical methods and computerised technologies used to process, extract and visualised data in order to detect patterns, relationships and insights that help businesses to make better decisions (IBM 2023).

Real World Example

→ Transnet National Ports Authority has implemented a SAP HANA - powered business analytics to measure cargo loading and unloading speeds across 8 South African ports.

→ The business analytics was implemented to reduce the reporting turnaround time from 8 days to 35 minutes.

② Data Analytics refers to a process of collecting, cleaning, transforming, analyzing and interpreting data in order to find meaningful patterns, trends and insights that will support informed decision making (www-coursera.org)

Real World Example

→ Transnet data analyst can analyze locomotive sensor data to predict equipment failures and schedule maintenance before costly breakdowns occur using Power BI, Python, SQL or databricks.

③ Business Intelligence (BI) refers to tools, infrastructure and practices that are used to collect, store and analyse business data for better decision making.

Real world example.

Transnet SAP business warehouse is a BI infrastructure that is used to collect port operational data and makes it accessible for analysis and planning.

④ Data-Driven Decision Making is a practice that business use to make decisions and business choices that are based on data analysis instead of data that is based on intuition or personal experience.

Real world example

Transnet Freight Rail had a problem of train delays on major freight corridors. The operational division (co) used data from train schedules, GPS tracking, locomotive availability, tonnes transported and network congestion to make an informed decision to improve efficiency and customer service.

⑤ Big Data refers to extremely large volumes of data that can be structured or unstructured, generated at a high speed from different or various sources.

Real world example

Transnet Freight Rail (TFR) manages thousands of trains, locomotives, wagons, signals, sensors, maintenance systems and operational records across South Africa.

On a daily basis TFR generates large volumes of data from GPS trackers on locomotives, Trackside sensors, Maintenance records, Train schedules, Fuel Consumption logs, Cargo tracking systems, weather data, port and terminal operations.

→ This data has millions of records generated daily, the data is streamed in real time from sensors and tracking devices.

→ The data is extracted from databases (Structured) logs (Semi-structured) and documents, images, videos (Unstructured).

⑥ Descriptive Analytics is a process of examining historical data through the identification of trends and patterns in the past business performance.

Real world Examples

Escom uses descriptive analytics through Power BI dashboards to visualise historical productivity and energy sales data spanning over 15 years from annual reports to understand past performance trends before the more strategic decisions.

⑦ Diagnostic Analytics is a process that investigates why events occurred by examining data to find root causes and relationships.

Real world examples

Escom uses artificial neural networks to diagnose the factors behind periods of growth and decline. The examination

③

is used to identify when specific managerial decisions correlated with improved or deteriorated organisational performance in order to help management and leadership understand causal relationships.

⑧ Predictive Analytics is used by businesses to forecast future outcomes and identify possible trends by using historical data, statistical models and machine learning.

Real world Analytics Example:

Tor-Snet National Ports Authority (TNPA) uses real time analytics to predict future trends across the 8 ports and this provides customers with insights optimal and suboptimal shipping times to reduce delays and supply chain disruptions.

⑨ Prescriptive Analytics is a process of analysing data and simulating scenarios in order to recommend specific actions that will help the business to achieve their desired outcomes.

Real world examples:

Esri uses neural network analysis to determine the moment when decisions should be taken to prevent the demise of the company, transferring predictive findings into strategic recommendations for management.

⑩ Data literacy refers to the ability to read, understand, analyse, interpret and communicate data in a meaningful way in order to make informed decisions. (www.gartner.com).

Real world example:

⇒ A manager in Transnet receives a dashboard showing that locomotives are delays has increased by 15% during the previous month.

⇒ A data-literate manager can understand the dashboard, the metrics presented, analyse the causes of delays, verify the source and reliability of the data, interpret the trends and patterns, communicate the findings and make evidence based decisions.

⇒ A manager without data literacy may rely on assumptions or past experiences rather than objective evidence.

Data, Measurement and Reporting

⑪ Structured Data is data that is organised, has a predefined format and normally arranged in rows and columns (IBM, www.ibm.com)

Real world example:

Employee ID	Employee Name	Department	Salary
100428	Lerato Mkhwanazi	Finance	R 99.000
121226	Donelle Greyling	ER	R 208.000
121224	Maguel Mdlouy	IT	R 25.000
131458	Nindla Mavlangu	HR	R 50.000
58342	Willem Britz	Catering	R 12.000

⑤

⑫ Unstructured Data is data that does not have a predefined format, is not organised in rows or columns and quite difficult to store, search and analyse (www.ibm.com).

Real world examples:

Videos

Emails

Chart images

CCTV Footages

Word or PDF documents

⑬ Data Warehouse is a centralised storage for structured, filtered data rearranged for analysing and reporting.

Real world examples:

Transnet has implemented SAP Netweaver Business Warehouse powered by SAP HANA to consolidate operational data from the 8 SA ports into a single analytical repository.

⑭ Data Mining is the process of discovering patterns and relationships in a large database using statistical and machine learning techniques.

Real world examples

Escom's analysis mined over 15 years of annual report data including annual productivity and energy sales using artificial neural networks to extract meaningful

patterns that would not have been evident through manual reviews.

(15) Data Visualisation refers to graphical representation of data using dashboards, charts and graphs to identify patterns and communicate insights.

Real world example:

Escom uses trained professionals in Power BI covered data visualisation, to create reports and dashboards that simplifies complex electrical data to ease the identification of past ~~unnecessary~~ results trends and patterns for more informed decision making.

(16) Dashboard is a visual interface displaying key metrics and KPI's in a form of charts, graphs and tables for quick insights.

Real world examples:

Escom uses Power BI dashboards to centralize and display energy sales and productivity metrics to allow management to monitor performance at a glance.

(17) Metric is a value that is measurable used to track, monitor and evaluate the performance of a process activity or business objective (Cloud.ibm.com)

Real world example:

Revenue, Operating Cost, Profit margins and ROI
A company tracks monthly revenue to assess financial

⑦

performance.

⑮ Key performance indicator (KPI) is a measurable value that indicates how an organisation is effectively achieving the key business objectives.

Real world experiences:

TINPA uses cargo loading and unloading speeds as a key operational KPI, which is measured through SAP business warehouse to maintain on-time efficiency across the 8 SA ports.

⑯ Benchmarking is a process used by organisations to compare products, services, processes and/or performance metrics against the industry leaders or best performing companies to identify gaps and opportunities for improvement (<https://asq.org/quality-resources/benchmarking>)

Real world examples

Suppose Transnet's annual training completion rate is 80% and the industry best practice organisation achieves 95%.

Metric	Transnet	Industry leader
Training Completion Rate	80%	95%
Pass Rate	85%	95%
Average Training for Days	10 Days	7 Days.

→ The learning administration team can identify performance gaps, learn best practices from top

performing organisations, set improvement targets and improve learner success rate and compliance levels through benchmarking.

② Segmentation is a process used to identify patterns, trends and opportunities by dividing and grouping data based on specific attributes.

Real world example?

Trennet wants to analyze training compliance in this instance:

Employee Group	Employees	Compliance Rate
Executives	100	98%
Managers	500	94%
Specialists	1,200	88%
Operations Employees	3,000	76%

Management can use segmentation to identify that operations employees require additional support and interventions.

Customer & Sales KPI's

② Conversion Rate is a metric that measures the percentage of people who completed a desired action out of the total number of people who had the opportunity to do so (<https://mailchimp.com/marketing-glossary/conversion-rate/>)

Real world example?

Suppose 500 employees visit Trennet learning portal and 125 employees successfully enroll in a training course.

⑨

$$\text{Conversion Rate} = \frac{125}{500} \times 100 = 25\%$$

~~This means that the conversion rate = 25%~~

This means that 25% of visitors completed the desired action which is the course enrollment.

② Customer Acquisition Cost (CAC) is a metric used by businesses to measure the total cost incurred to acquire new customers (<https://mailchimp.com/marketing-glossary/conversion-rates/>).

Real life example

Suppose a company spends

R50 000 on marketing

R30 000 on sales team salaries

R20 000 on Advertising

∴ The total acquisition cost = R100 000

If the company acquires 200 new employees during the same period

$$\text{CAC} = \frac{\text{R100 000}}{200} = \text{R500}$$

CAC = R500 per customer.

This means the company spent R500 to acquire a new customer.

③ Customer Lifetime Value (CLV/LTV) is a metric used to estimate the total revenue or profit a customer is expected to generate for a business over the entire duration of their relationship with the organisation. (Kotler, P, Keller, K.L & Cherneva, A 2022, marketing management (16th ed). Pearson.)

Real life example

Suppose a customer

→ Spends R2000 per purchase

→ Makes 4 purchases per year

→ Remains a customer for 5 years

$$CLV = R2000 \times 4 \times 5$$

$$CLV = R40000$$

This means the customer is expected to generate R40 000 in revenue over their lifetime with the company.

② Churn Rate is a metric used to measure the percentage of customers, subscribers, employees or users who stop using a product, service or program during a specific period.

Real world example

A company starts the month with 1000 customers at the beginning, during the month 50 customers are lost (Kotler, P, Keller, K.L. & Chernev, A. (2022) Marketing Management (16th ed.) Pearson.

$$\text{Churn Rate} = \left(\frac{50}{1000} \right) \times 100$$
$$= 5\%$$

This means 5% of customers stopped doing business with the company during the month.

② Customer Retention Rate is a metric that measures the percentage of customers who continue doing business with an organisation over a specific period. (Ibid)

Real world example

= A company has 1000 customers at the beginning of the year.

= New customers acquired during the year are 300

= At the end of the year the company has 1150 customers.

$$\begin{aligned} CRR &= \left(\frac{1150 - 300}{1000} \times 100 \right) \\ &= \left(\frac{850}{1000} \times 100 \right) \\ &= 85\% \end{aligned}$$

This means the company's retention rate is 85% of its existing customers during the year.

② Net Promoter Score (NPS) is a customer loyalty metric used to measure the likelihood customers are to recommend a company's products, or services to other people. (ibid)

Real world example^a

A company uses a survey to obtain data from customers; in this instance 100 customers.

Promoter (9-10): 60

Passive (7-8): 25

Detractors (0-6): 15

$$\% \text{ Promoters} = \frac{60}{100} \times 100 = 60\%$$

$$\% \text{ Detractors} = \frac{15}{100} \times 100 = 15\%$$

$$\begin{aligned} \% \text{ NPS} &= 60\% - 15\% \\ &= 45\% \end{aligned}$$

This indicates a strong level of customer loyalty

⑫

②⑦ Average Order Value (AOV) this is an e-commerce and Sales metric that measures the average amount of money customers spend per order or transaction. (Ibid)

Real world example?

Suppose a ~~Customer~~ Company generates R500 000 in sales from 2000 order during a month.

$$\begin{aligned} AOV &= \frac{R500000}{2000} \\ &= R250 \end{aligned}$$

This means customers spend an average of R250 per order.

②⑧ Sales Funnel refers to a visual form model that represents the Customer's journey from first becoming aware of a product or service to making a purchase

Real world examples?

Awareness	10,000
Interest	2000
Consideration	800
Intent	500
Purchase	350

Conversion Rate between stages.

$$\begin{aligned} \text{Interest Rate} &= \frac{2000}{10000} \times 100 \\ &= 20\% \end{aligned}$$

$$\text{Consideration Rate} = \frac{800}{2000} \times 100 \\ = 40\%$$

$$\text{Purchase Rate} = \frac{350}{10000} \times 100 \\ = 3.5\%$$

This analysis helps to identify where potential customer or learners drop out of the process.

Marketing & Campaign Analytics

②⑨ Marketing Campaign is a set of marketing activities implemented to accomplish a particular marketing objective, such as introduction of new products, building brand awareness etc. (Standard bank Newstram, 2026).

Real world Example

Standard bank's "money reels" is a year-end customer account overview campaign that was personalised and turned into videos for every customer. As a result, there were nearly half a million customers and 537 new app installations per bank.

③⑩ Post-Campaign Analysis is the process of evaluating the success and short comings of a completed marketing campaign, to identify opportunities for improvement and guide future decisions. (Standard bank Newstram 2026)

Real world examples:

Standard bank conducts an analysis called the post-campaign

Real world examples:

Standard bank conducts a post-campaign analysis for the Money Reels Campaign. The analysis indicated that the engagement resonated across generations and almost a quarter of the audience was over the age of 50 years. This assessment is expected to influence the bank to adopt the format beyond the year-end customer recap.

(31) Incremental Revenue is the additional revenue derived from an extra unit of sales, over and above the amount expected if the campaign had not been launched (Shoprite Holdings Integrated Report 2024).

Real world examples:

In Shoprite, the extra savings programme accrues R16,9 billion in discounts to its customers.

⇒ However, it is a source of incremental revenue for the group as its customers shop more frequently and with large baskets.

⇒ According to the report 85% of the total sales at Shoprite and Checkers are now from loyalty programme customers.

(32) Return on Investment (ROI) measures the performance of an investment, it is calculated as the ratio of money gained or lost from investment compared to the amount of money invested (Plexera SASOL Case Study).

Real World Example:

In the SaaS software asset management case, the investment paid for itself many times over, with savings and avoidance of \$34 million in costs.

⇒ Although it is not a marketing campaign, this is an illustration of return on spend.

③ Return on Ad Spend (ROAS) is a metric that measures how much revenue is generated for every ~~rand~~ invested in digital marketing campaigns (Moneyweb 2024).

Real World example:

⇒ Choccos Sixty6's R99 monthly subscription offering is designed to encourage recurrent orders as a function of the R35 delivery charged on every order.

⇒ The value proposition implied by the shift in pricing is likely to improve the economics of every delivery for the company as it continues to invest in ~~advertising~~ advertising and promotions.

④ Click through Rate (CTR) refers to the percentage of users who clicked on a link compared to the number of times an ad is displayed (Standard bank NewRoom 2026)

16

Real world example:

⇒ Standard bank's Money Reels campaign recorded an eye-popping 80% completion rate.

⇒ Although it is not a click-through rate, the bank

16

can presume that the percentage of customers who decided to watch the video rather than skip it to the next slide is significant, ~~based~~
→ Besides the nearly half a million customer engagement, the campaign recorded 837 new app installations.

(35) Bounce rate is the percentage of website visitors who leave after viewing only one page.

Real world examples

Standard bank uses mining to identify journey variants correlated with increased completion time, which probably indicates where customers bounced during registration.

(36) A/B Testing is a method used to compare two versions of a web page to determine which one produces better results (moneyweb 2020)

Real world examples

Choccos personalisation of CityGo offers for its "just for you" segment on its mobile application is a result of extensive A/B testing in identifying which offers resonate with which customers.

(37) Lead Generation is a process of capturing the attention of prospective customers and persuading them to get in touch with a company.
(Shoprite Holdings Integrated Report 2024)

Real world examples:

Shoprite use Xtra Savings programme as a lead-generating campaign.

→ Every customer who applied for a card provides the bank with personal information that is used to market to them more personally.

③8 Marketing Attribution is a process used to to identify and analyse which marketing ^{channel or} campaign elements is responsible for driving conversions or sales. (ARIS/Standard bank case study 2026).

Real world examples:

Standard bank's observability stack enabled the bank to visualise what was happening in the systems live and begin to understand the impact of the various digital interventions on its customers.

References

- 1) www.ibm.com
- 2) Cloud.ibm.com
- 3) www.coursera.org
- 4) www.gartner.com
- 5) www.transnet.co.za
- 6) Transnet. (2025). Annual Performance Report. Transnet SOC Ltd.
- 7) Shiftr. (2025). Azure Power BI Eskom. .
- 8) Eboule, P. (2023). Decision-making based on machine learning techniques: A case study of Eskom. Polish Journal of Management Studies, 28(1). .
- 9) Flexera. (n.d.). Sasol drives environmentally aware growth via \$34M cost savings. .
- 10) ARIS. (2026). How Standard Bank Transformed Process Efficiency with ARIS. .
- 11) TechCentral. (2024). How Shoprite is using AI to sell you more groceries. .
- 12) Moneyweb. (2024). How Sixty60 is turning Checkers into an unbeatable machine.
- 13) <https://asq.org/quality-resources/benchmarking/>
- 14) <https://mailchimp.com/marketing-glossary/conversion-rate/>
- 15) SAP. (n.d.). Sasol Case Study: Global Management of Chemical Division. .
- 16) Teradata. (2024). Omni-Channel Personalization at Standard Bank. .
- 17) Standard Bank. (2026). When data tells an impactful story: Standard Bank's Money Reels earns top industry honour. .
- 18) Eskom Data Scientist Presentation (Jaco du Toit). (n.d.). ESI Africa. .
- 19) Shoprite Holdings. (2024). Integrated Report for the year ended 30 June 2024.
- 20) Kotler, P, Keller, KI & Chernev, A 2022, marketing management (16th ed.) Pearson
- 21) Flexera/Sasol case study